

Digital Platform for Multimodal Visibility of Coal Transportation

A 100% Free-Tier Cloud Platform Providing End-to-End Tracking, Groq LLM & XGBoost Delay Prediction, and Multimodal Decision Intelligence

Project ID:	289 (CSE7102)	Target Release:	Version 1.0 (Nov 2026)
Authors / Team:	Manaswi Bhujange (20231CSE0430) Shinde Nandini (20231CSE0421) Biradar Ashwini (20231CSE0400)	Supervisor / Guide:	Prof. Nayana R, Dept. of CSE Presidency University, Bengaluru
Primary Tech Stack:	React.js, Spring Boot, MySQL, Python, Groq API (Free), Leaflet/OSM	Cost & Tier Policy:	100% Free-Tier & Open Source Zero API billing / No credit card lock-in

1. Executive Summary & Problem Context

Coal fuels approximately **70% of India's commercial electricity generation** (Ilahi, 2024). The supply chain operates through a complex multimodal network connecting coal mines (e.g., Jharia, Korba, Singrauli) via road tipper trucks to railway sidings, loading onto bulk rakes/wagons, and concluding at pit-head or distant thermal power plants. Despite its critical national significance, the logistics chain remains severely fragmented.

Core Problem Statement: Coal shipment status is fragmented across disconnected road and rail logistics systems. Stakeholders lack real-time visibility, leading to delayed disruption detection, severe rake turnaround delays (~5 days), sub-optimal siding loading, and high turnaround penalties. Decision-makers operate with partial or stale information, resulting in fuel shortages at power plants and stranded coal reserves at mines.

Key Industry Metrics (Empirical Basis):

- Stranded Coal:** ~60 Million tonnes of coal was stranded at Indian mine pitheads in 2022 due to evacuation bottlenecks.
- Rake Utilization Deficit:** Actual loading reached only 241 rakes/day against a national target of 273 rakes/day (Ministry of Coal, 2022).
- Turnaround Delays:** Indian railway wagon turnaround averages ~5 days, driven by idle siding dwell times and blind intermodal transitions.
- Operational Disruptions:** Over 1,900 passenger trains were cancelled or diverted in peak crisis periods to prioritize coal transport corridors.

2. Product Objectives & Target KPIs

The digital platform bridges the information divide across road and rail logistics stages through real-time tracking, automated predictive modeling, and role-tailored decision support under a **100% zero-cost software architecture**.

Objective	Target Metric / KPI	Measurement Method
Multimodal Unification	100% of road legs (mines to sidings) and rail legs (sidings to power plants) mapped on a single dashboard.	Platform consignment audits and stage mapping verification.
ETA Accuracy	Predicted arrival times within ±45 minutes of actual arrival across a 24-hour transit horizon.	Mean Absolute Error (MAE) evaluated on historical & live trips.
AI Disruption Insights	Zero-cost automated reasoning on delay causes & mitigations via Groq API (LLaMA 3.3) under < 1.5s latency.	Groq API token execution and response validation benchmarks.
Information Latency	Live tracking map and status updates refreshed in < 5 seconds; backend API latency < 200 ms.	System health telemetry and synthetic load testing.
Zero-Cost Operation	Zero monthly billing across all components (Leaflet/OSM, Groq Free Tier, Open-Meteo, MySQL).	Monthly expense audit confirming \$0 cloud API expenditures.

3. Target Personas & Role-Based Access Control

The platform enforces strict Role-Based Access Control (RBAC) via Spring Security and JWT tokens. Each user profile accesses customized views aligned with their operational duties:

Role	Primary Responsibilities	Key Features & Permissions
Logistics Manager	Oversees end-to-end supply chain from mine to power plant.	Full fleet dashboard, aggregated KPI analytics, delay probabilities, consignment creation, Groq AI root-cause briefings.
Operations Team	Manages daily dispatch, siding loading, rake scheduling, and unloading.	Real-time live map, active alert feeds, exception acknowledging, manual event updates, siding dwell time logs.
Transporter / Fleet Owner	Operates contracted truck fleets and rail rakes.	Vehicle route visibility, driver assignment, speed logs, route deviation alerts, trip completion check-ins.
System Administrator	Platform infrastructure, API gateway, integrations, and user lifecycle.	User account management, role assignment, audit logs, Groq API key setup, model retraining pipeline triggers.

4. Functional Requirements & Module Specifications

The system architecture comprises six interconnected functional modules supporting the complete transit lifecycle.

Module 1: Authentication & User Management (AuthService, UserService)

- **FR-1.1 Secure Authentication:** Login via username/password, issuing cryptographically signed JWT bearer tokens with configurable TTL (8 hours).
- **FR-1.2 Role Authorization:** Enforce granular endpoint permissions based on user roles (Admin, Logistics Manager, Operations, Transporter).
- **FR-1.3 Profile Management:** Admin panel to create, update, suspend, and assign organizations/regions to user accounts.

Module 2: Multimodal Consignment & Shipment Management (ShipmentService)

- **FR-2.1 Consignment Lifecycle:** Register coal shipments capturing Consignment ID (e.g. CL-78211), coal grade, tonnage (e.g. 3,500 MT), mine (Jharia Coalfield), siding (Raipur Terminal), and power plant (Kawardha).
- **FR-2.2 Multimodal Leg Segregation:** Support multi-leg itineraries (Leg 1: Road via Truck, Leg 2: Rail via Rake R-1256, Leg 3: Final Mile Road).
- **FR-2.3 State Machine:** Maintain strict consignment statuses: PLANNED, DISPATCHED, IN_TRANSIT, SIDING_HANDOVER, DELAYED, DELIVERED, CANCELLED.

Module 3: Live Multimodal Tracking & Geospatial Visualization (TrackingService, RouteService)

- **FR-3.1 Free Open-Source Map Engine:** Vector map built on Leaflet.js + OpenStreetMap (OSM) tiles (100% free, replacing paid Google Maps API) with zero billing overhead.
- **FR-3.2 Free Open Routing:** Route calculation & waypoints powered by OSRM (Open Source Routing Machine) and Open-Meteo for live weather data without API billing keys.
- **FR-3.3 Telemetry & Breadcrumbs:** Moving markers showing coordinates, speed, heading, and *Latest Known Status* with a 'Signal Stale' badge upon network dropouts.
- **FR-3.4 Geofence Triggers:** Configurable geofences around mine gates, weighbridges, and siding junctions with automated milestone timestamps.

Module 4: AI/ML Predictive Analytics & Groq LLM Intelligence Engine

- **FR-4.1 Quantitative ETA & Delay Model:** Supervised regression & classification (XGBoost / Scikit-learn) estimating numerical arrival timestamps and delay probabilities (0–100%).
- **FR-4.2 Groq API Generative Decision Support:** High-speed free-tier Groq API integration (LLaMA 3.3 70B / Mixtral) generating natural language root-cause explanations and mitigation recommendations for delays.
- **FR-4.3 Operational Intelligence Summaries:** Groq LLM synthesizes telemetry exceptions, weather data, and siding bottlenecks into instant bulleted action briefs for Logistics Managers.
- **FR-4.4 Spatial Anomaly Detection:** Flags unapproved corridor deviations > 5 km or stationary siding bottlenecks exceeding 45 minutes.

Module 5: Exception & Alert Notification Engine (AlertService, NotificationService)

- **FR-5.1 Alert Categorization:** Classify alerts into High Severity (route deviation, delay > 2 hrs, signal halt), Medium (traffic, speed violation), and Info (geofence milestones).
- **FR-5.2 Real-Time Broadcast:** Dispatch instant notifications via WebSocket to the dashboard notification tray with auditory and visual flags.
- **FR-5.3 Free Notification Dispatch:** Alert summaries synthesized by Groq API and delivered via free SMTP email gateways (Gmail/Resend free tier).
- **FR-5.4 Acknowledgment Workflow:** Operators can acknowledge alerts, add resolution notes (e.g., 'Locomotive change in progress'), and mark resolved.

Module 6: Executive Analytics, Reports & Dashboard (AnalyticsService, ReportService)

- **FR-6.1 Summary KPI Cards:** Real-time metric cards showing Total Active Shipments, In Transit, Delivered, and Delayed counts with monthly % trends.
- **FR-6.2 Fleet Mode Distribution:** Graphical breakdown of active consignments across Road (35%), Rail (55%), and Combined Multimodal (10%).
- **FR-6.3 Delay Bucket Breakdown:** Donut chart categorizing shipments: Normal (64%), Delayed < 6 Hours (25%), Delayed > 6 Hours (11%).
- **FR-6.4 Automated Report Export:** One-click generation of PDF / Excel audit reports on turnaround times, rake demurrage penalties, and carrier SLA compliance.

5. Non-Functional Requirements (NFRs)

Category	Specification / Requirement	Standard / Validation
Performance	API response time < 200 ms for 95th percentile. Dashboard initial load < 2.5 seconds. Groq API inference latency < 1.5 seconds.	JMeter load test under 500 concurrent requests.
Scalability	Horizontal scaling of Spring Boot backend containers to support up to 5,000 active consignments and 20,000 telemetry events/minute.	Stateless microservices with auto-scaling triggers.
Security	End-to-end TLS 1.3 encryption. Passwords hashed using BCrypt. OAuth2 / JWT authentication. Secure API key storage in environment variables.	OWASP Top 10 security compliance audit.
Reliability & Availability	System uptime ≥ 99.5%. Automated database backups every 24 hours. Graceful offline cache fallback for intermittent mobile networks.	Cloud SLA & health probes with automatic container restarts.
Cost & Portability	100% Zero-Cost Free-Tier stack: No credit card requirement for Google APIs. Complete operational self-sufficiency on free community tiers.	Monthly billing verification (\$0 total cost).

6. System Architecture & Free-Tier Technical Stack

The architecture is engineered to run completely on free-tier, open-source services without paid API lock-in:

Architectural Tier	Technology / Framework	Free-Tier & Zero-Cost Specification
Presentation Tier	React.js (v18+), TailwindCSS, Leaflet.js	Open-source frontend framework. Hosted on Vercel / Netlify free tier with zero ongoing hosting fees.
Mapping & Geospatial	Leaflet.js + OpenStreetMap (OSM) / OSRM	Replaces Google Maps API: Free public tile servers, no billing account, no credit card required, unlimited rendering.
API & Application Tier	Java 17, Spring Boot 3.x, Spring Security	Open-source Java framework; modular REST architecture; JWT authentication; runnable locally or on Render free tier.
AI / ML Intelligence Tier	Python 3.9+, XGBoost + Groq Cloud API	Groq Free Tier (LLaMA 3.3 70B / Mixtral) for lightning-fast NLP disruption reasoning + Local XGBoost/Scikit-learn for ETA.
Weather & Route APIs	Open-Meteo API & OSRM Engine	Free weather data (10,000 calls/day, no key) and free road routing engine replacing paid Google Directions APIs.
Data Storage Tier	MySQL 8.0 Community Edition	Open-source relational database running locally or on free cloud tiers (Aiven / Neon / Railway free tier).

7. Data Model & Database Schema Specifications

The relational schema in MySQL is organized around high-integrity operational tracking:

Entity	Key Attributes / Fields	Relationships & Constraints
users	user_id (PK), username, email, password_hash, role, organization, created_at	Enumerated role: LOGISTICS_MGR, OPS, TRANSPORTER, ADMIN.
shipments	shipment_id (PK), consignment_code, coal_grade, tonnage, origin_mine, destination_plant, current_mode, status, eta, groq_summary, created_at	1-to-many with shipment_legs and alerts; stores Groq AI incident briefing.
shipment_legs	leg_id (PK), shipment_id (FK), leg_sequence, mode (ROAD/RAIL), carrier_name, vehicle_or_rake_no, start_node, end_node, status	Maintains sequential transit order and multi-modal transitions.
telemetry_logs	log_id (PK), leg_id (FK), latitude, longitude, speed, heading, timestamp, signal_status	High-frequency positional updates; indexed on (leg_id, timestamp).
alerts	alert_id (PK), shipment_id (FK), severity, category, message, ai_recommendation, is_resolved, resolved_by, created_at	Includes Groq LLM recommended mitigation action for fast operator resolution.

8. Implementation Roadmap & Milestones

Structured in alignment with the academic project milestones (Aug – Nov 2026):

Phase & Date	Key Deliverables & Objectives	Review Milestone
Phase 1 08 Aug – 28 Aug 2026	Literature review of 16 research papers; problem statement definition; PRD compilation; baseline tech stack selection.	Review 0 (Completed)
Phase 2 29 Aug – 21 Sep 2026	System architectural design; UI wireframes; database ER modeling; prototype scoping (MySQL + Spring Boot + React).	Review 1 (Completed)
Phase 3 22 Sep – 20 Oct 2026	Development of Spring Boot REST APIs; JWT security layer; React dashboard layout; Leaflet + OpenStreetMap integration.	Review 2 / Check
Phase 4 21 Oct – 09 Nov 2026	AI/ML model training (XGBoost); Groq API integration for disruption analysis; alert rule engine & Open-Meteo weather.	Feature Freeze
Phase 5 10 Nov – 20 Nov 2026	End-to-end multimodal workflow validation; synthetic load testing; Docker containerization; cloud deployment & viva documentation.	Final Demo & Submission (20 Nov 2026)

9. Key Risks, Assumptions & Free-Tier Mitigations

Risk / Challenge	Severity	Free-Tier Mitigation Strategy
Paid API Costs (Google Maps / Cloud)	High	Completely eliminated: Built on Leaflet.js + OpenStreetMap and OSRM routing; zero credit-card dependency and no usage invoices.
Groq API Rate Limits on Free Tier	Medium	Implement in-memory caching (Redis/Spring Cache) for LLM disruption summaries and throttle calls to trigger only upon major exception events.
Restricted Railway FOIS API Access	High	Emulate FOIS / ULIP railway event data structures using realistic rail siding handover datasets derived from Ministry of Coal published schedules.
Intermittent Cellular Network on Corridors	Medium	Dashboard implements a 'Latest Known Position' state with visual timestamp freshness indicators, preventing system crashes upon packet loss.

10. Key References (Literature Survey Alignment)

1. Ministry of Coal, Govt. of India (2022). *Smart Coal Logistics Plan (SCLP) & Draft National Coal Logistics Plan*. Emphasizes integrated multimodal freight visibility.

2. Ilahi, M. (2024). *Coal supply chain in India: Challenges & road ahead*. TIJER. Empirical baseline for 70% electricity reliance and 60 MT stranded coal pithead inventory.

3. Pandey et al. (2025). *Multi-modal visibility of coal transportation*. ICAAAI 2025, Atlantis Press. Architecture and operational paradigm for unified coal visibility platforms.

4. Balster et al. (2020) & Wani (2025). *Arrival-time prediction for intermodal networks & ML ETA improvements*. PeerJ Computer Science.

5. Kalaïarasan et al. (2022). *The ABCDE of supply chain visibility*. Int. J. Production Economics.

Document Sign-off & Academic Approvals: This Product Requirements Document establishes the technical baseline and functional scope for CSE7102 Mini Project implementation at Presidency University. Engineered for 100% free-tier operation (Groq API, Leaflet/OSM, MySQL). Approved by the student team (Manaswi Bhujange, Shinde Nandini, Biradar Ashwini) under Prof. Nayana R.